

PROJECT Smart Queue Management System



Less
Waiting Time



Better
Organization



Improved
Efficiency



NAME

R.THENMOZHI



COLLEGE

RVS SOET



DEPARTMENT

B.E / CSE



Smart Queue Management System

A Smarter Way to Manage Your Queue



Problem Statement

- Traditional queue management systems require people to wait for a long time and may cause overcrowding and confusion.
- Manual queue handling also makes it difficult to manage customers efficiently and provide timely service.
- Therefore, an intelligent system is needed to manage queues efficiently, reduce waiting time, and improve the overall user experience.



Brief Description of the Project

The Smart Queue Management System is a web-based application designed to automate and simplify the queue process in service-based environments such as banks, hospitals, government offices, and customer service centers.

The system allows users to get a token, track their position in real-time, and receive notifications when it's their turn. It helps reduce waiting time, prevents overcrowding, and improves the overall service efficiency.



Get Token

User collects a token number



Track Position

View live queue status



Get Notified

Receive alert when it's your turn



Receive Service

Visit the counter and get served

Project Objectives

Our goals for a smarter and more efficient queue system.



To reduce customer waiting time.



To manage queues efficiently.



To reduce overcrowding and confusion.



To provide real-time queue status.



To improve the overall service experience.

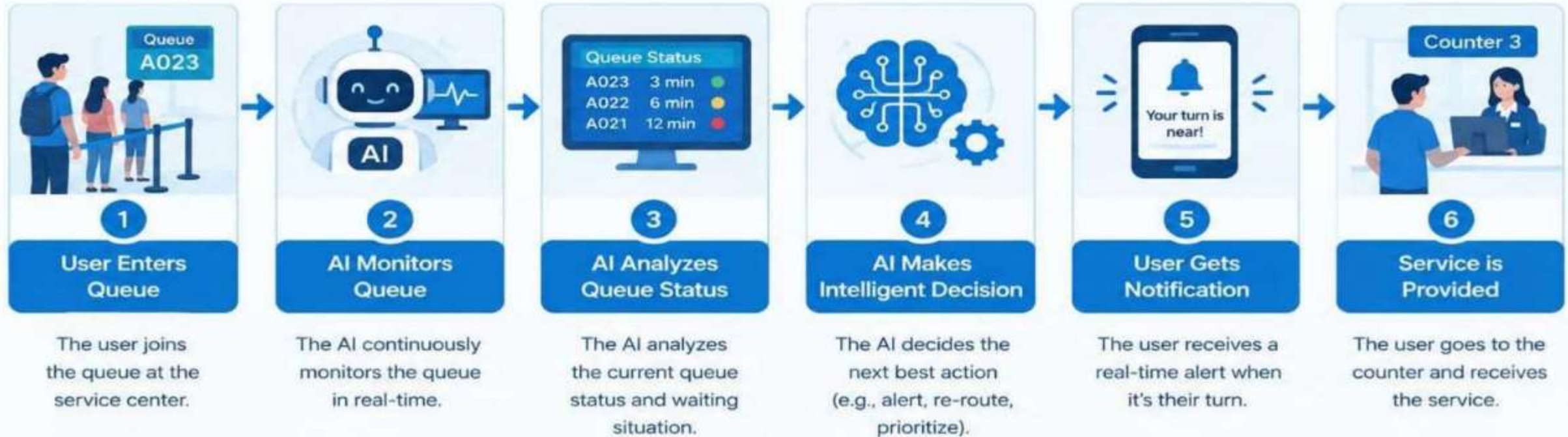


To automate the queue management process.



How the **Agentic AI** Solution Works

From Queue to Service – Smarter, Faster, Easier



05 | Key Features

Smart Queue. Smarter Service.



Digital Token Generation

Get your queue token easily.



Real-Time Queue Tracking

Track your position live.



AI-Based Queue Monitoring

Monitor queues intelligently.



Turn Notification

Get notified when your turn comes.



Crowd Management

Manage crowd efficiently.



Reduced Waiting Time

Minimize unnecessary waiting.



Automated Queue Management

Automate queue operations.



User-Friendly Interface

Simple and easy to use.

06 | Technologies / Tools Used

Building the solution with the right tools.



Python

Used for building the backend logic, data processing and AI integration.



VS Code

Used as the code editor for writing and managing the project files.



07

Working Process

From queue entry to service – Powered by AI.



08 | Screenshots / Output



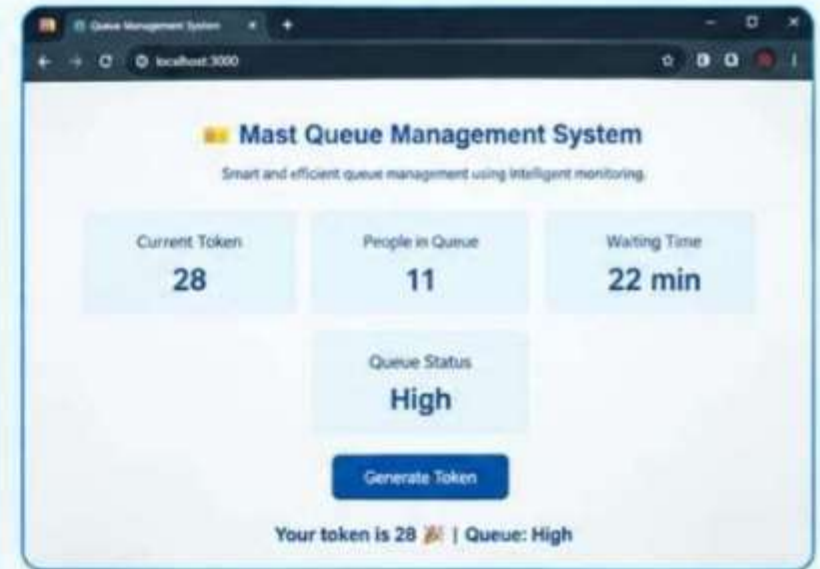
 Screenshot 1 — Main Queue Dashboard

- Current Token
- People in Queue
- Waiting Time
- Queue Status



 Screenshot 2 — Token Generation

- Generate Token button click
- Token number update
- Queue count update
- Waiting time update



 Screenshot 3 — High Queue Detection

- People > 10
- Queue Status: High

09 | Results Achieved



Reduced manual
queue management



Waiting time can be
estimated dynamically



Real-time queue count
is displayed



Tokens are generated
automatically



Queue status is monitored
using intelligent logic



User-friendly
queue interface



Faster and more organized
queue management

10 | Conclusion



The system provides a simple and efficient way to manage queues.



Digital tokens help organize customers in a systematic manner.



Queue count and waiting time are updated dynamically.



Intelligent queue monitoring helps identify high queue conditions.



The system can reduce confusion and improve the overall user experience.



It provides a foundation for a future AI-powered queue management system.



Smart Queue Management for Faster and Better Service

11

Challenges Faced



Setting up the development environment



Learning Python and Flask integration



Connecting the frontend with backend



Updating token and queue data dynamically



Implementing intelligent queue-status logic



Testing and fixing errors during development



Challenges helped improve our technical and problem-solving skills.

12

Future Enhancements



AI-Based Prediction

Predict crowd levels and waiting time using machine learning.



Mobile Application

Allow users to get tokens, check queue status and notifications on mobile.



Cloud Deployment

Host the system on cloud for better scalability and performance.



Smart Notifications

Send real-time SMS / app notifications when it's your turn.



Advanced Analytics

Provide reports and insights for better decision-making.



Better Security

Add authentication and data protection for safe and reliable usage.



More features. Smarter system. Better experience.



Python Documentation

Official Python documentation for language reference and concepts.

<https://docs.python.org/3/>



Flask Documentation

Official Flask documentation for web framework and API development.

<https://flask.palletsprojects.com/>



Visual Studio Code

Official VS Code documentation for development environment setup and extensions.

<https://code.visualstudio.com/>



W3Schools

HTML, CSS, JavaScript and web development tutorials.

<https://www.w3schools.com/>



Project Reference

Smart Queue Management System – Concept, Design and Implementation (Online Resources).

<https://www.github.com/> (for similar projects)



Additional Learning Resources

GeeksforGeeks, TutorialsPoint, MDN Web Docs (for additional concepts and reference).

<https://www.geeksforgeeks.org/> | <https://developer.mozilla.org/>



References help build a stronger foundation
for better learning and future improvements.



THANK YOU

Mast Queue Management System

Smart Queue Management for Faster and Better Service



Presented By:
R.THENMOZHI



Save Time



Better
Management



Less
Crowd



Happier
Customers

